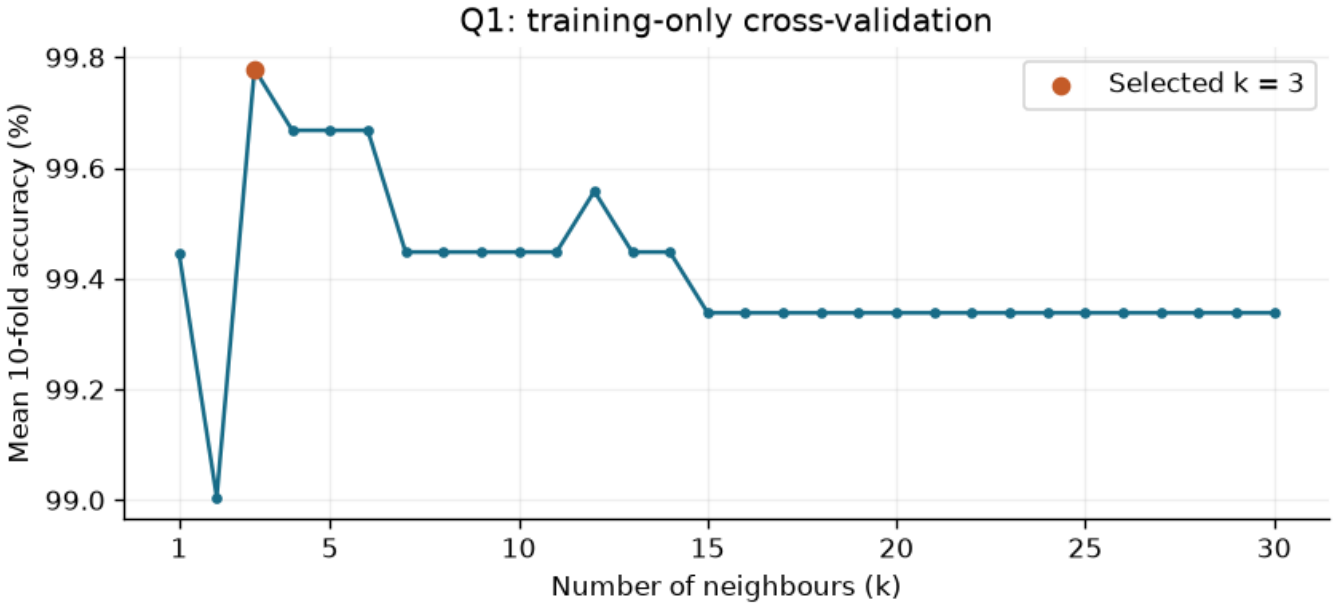
Q1: K-nearest neighbours

Method

The training set contains 904 images, each with 64 patch-count features and a label of 4 or 9. Euclidean distance is calculated as sqrt(sum((x - z)^2)). The label is excluded. An explicit vote count predicts the most common label among the k nearest neighbours. No built-in KNN classifier or distance function is used.

All k values from 1 to 30 are evaluated using the same stratified 10 folds, shuffled with random_state=42. Scores are the means of the ten validation accuracies. Vote ties choose 4, equal distances retain row order, and tied CV scores choose the smaller k. All features share the 0-16 scale, so no scaling is applied.

Cross-validation



Best k: 3
Mean cross-validation accuracy: 99.7790%

Test result

After selecting k, all 904 training examples are used as the reference set. The 226 test examples are loaded and evaluated once, after model selection.

Test accuracy: 99.5575% (225/226 correct).

Only one test digit is misclassified. This result measures classification of 4 and 9, rather than all ten digits.

Data: supplied knn-dataset/train.csv and test.csv.